Introduction:

The Philippines is a country with 99.9% dependency on foreign manufactured and developed medical equipment. In the Philippines talent is not the problem the framework and support structure is. This leads to:

- Delayed Translation to Clinical Use: Without a clear framework, innovators often lack regulatory guidance during early research and development (R&D), making it difficult to translate new technologies into clinical care.
- Stifled Innovation and High Costs: Fragmentation results in repetitive and costly submission requirements. Smaller firms or academic researchers may struggle to advance devices past the ideation stage due to the "medical device lag" caused by unharmonized standards.
- Compromised Data Quality and Objectivity: The lack of a shared evaluation framework makes it difficult to compare device efficacy objectively, as clinical and evaluation data are often not standardized or shared.
- Ethical and Safety Concerns: Missing policies can lead to ethical issues in human trials, particularly in regions where researchers may not have clear oversight or participants may not be fully informed.
- Unreliable Product Lifecycles: Without a product lifecycle management framework, up to 70% of medical equipment in resource-poor settings fails because of design barriers, such as a lack of required consumables or technical expertise that should have been addressed during the development phase.

The ISOXIAN Framework is a process that breaks development and commercialization into 21 different sections that can be used to establish a comprehensive Medical Device Development Lifecycle. This is a working manual that integrates international standards like ISO 13485, ISO 14971, ISO 62366 with specific Philippine regulatory requirements, and as well as the Technical Standard for Medical Device Management AS/NZ 3551 processes.

The framework outlines a structured six-phase progression covering project initiation, rigorous engineering and risk management, and the execution of human clinical trials. It defines critical organizational roles, such as the Quality Management Officer and Product Manager, while mandating strict document control and traceability through a "Golden Thread" of records. Technical guidelines emphasize cybersecurity by design, local EMC compliance, and a robust computer support infrastructure for managing documents, feedback and software anomalies.
Additionally, the documents provide detailed protocols for post-market surveillance, corrective actions, and the safe decommissioning of equipment. Ultimately, this collection serves as a definitive operational guide for ensuring safety, legal compliance, and quality throughout the entire lifespan of a medical product.

The full document can be accessed through a letter requesting access sent to the Institute of Biomedical Engineering and Health Technologies that holds the original copies of the 22 sections. This version is the concise version of the framework meant to initiate users into the framework while the 22 sections allow readers to see some details of the process. More detailed information can be found in the ISO 13485, ISO 14971, ISO 62366 standards that were customized by Voltaire Dupo for the APRO UX Study which is the input preparation part of the pre-commercialization process for APRO.

This is dedicated to the memory of my father Henry C. Dupo, an independent researcher that did work submitted to the Fertilizer and Pesticide Authority of the Philippines for AgroChemical Companies needing third party tests to verify efficacy of their products. This allowed me the opportunity to take up and finish engineering school. He also allowed me to participate in the process of gathering and interpreting data allowing me to learn another area past engineering. This showed me a valid framework for doing research and work is an essential part of getting the job done right each time-every time.

Voltaire B. Dupo 18 April 2026
Calamba, Laguna, Philippines

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ISOXIAN Framework Section Summary

SECTION A: RESEARCH AND DEVELOPMENT (PRODUCT DESIGN)

A0: Key Roles and Positions

The framework defines management leadership (Director for Academic Research Centers/ CEO for business startups/ spin offs and the CTO) responsible for setting the Quality Policy and signing the Declaration of Conformity. A Person Responsible for Regulatory Compliance (PRRC) remains independent to oversee technical release and vigilance should be onboard during the initial discussions for the project. A Business Consultant and Subject Matter Expert should be onboard to handle questions related to market needs, market fit and market research.

A Quality Management Officer (QMO) ensures QMS documentation is maintained. This can be internal (hired) or externally contracted by the project upon start of project.

Technical leadership includes a Product Manager for documentation and a Project Leader to head the medical team.

A1: Project Initiation & Administrative Setup

Projects begin with a Management Decision and the appointment of leadership.

For R&D Project Stage:

The Director of the Academic Research Center should work with the PRRC, Business Consultant and Subject Matter Expert to determine the use of the device, market need, market size and market acceptability requirements. This would define the project R&D objectives and build the proposal documents based on real world data and requirements.

For Startup/Spin Off - License Creation Stage

Data from the R&D project such as Customer Discovery, Market Research, Market Fit Evaluation, Focus Group Discussions, Value Engineered product, Industrial Design Patents,

Patents Full and Utility, Design Manufacturing File, Software/Firmware Audit Results, PSL Results and CISPR Evaluation Results and initial UX Study Results are used to create a Product Dossier that can be sold to willing parties to license the product.

Startup/Spin Off Launch Stage

Startup/Spin Off must outline the plan to:

1. Look for partnerships and networks to obtain the required medical collaborator groups or persons to launch clinical studies to validate the effectiveness of the device once the Class A License has been granted.
2. Manufacture the Product
3. Distribute the Product
4. Monitor the Product once launched
5. Provide Support for the Product once launched

Upon completion of License Negotiation and transfer of Product Dossier the Startup/Spin Off can start the process to File an FDA Class A License for Clinical Evaluation. Plans 1 to 5 must be re-evaluated upon receipt of the Dossier to check for validity of assumptions and other issues not seen until the Dossier File was transferred.

In Service Life

Startup / Spin Off must provide after Sales Monitoring and tracking of product for post market surveillance purposes. It is recommended that its Service Unit should create a technical management system that utilizes a 7-column Kanban board (Pre-Purchase to Decommissioning) for lifecycle tracking.

A2: Engineering, Architecture, and Verification

This phase involves creating a Software Development and Maintenance Plan and a Software System Test Plan. Developers use an evolutionary strategy for architecture design, coding, and unit testing. Verification includes formative usability evaluations to ensure the user interface meets design inputs. Items such as Software Testing related to Reliability and Stability the Framework uses ISO 62304 but audit items are related to MISRA C++ for embedded firmware evaluation.

A3: Document and Record Control

Documents are managed in "drafts," "under review," and "released" folders. A specific naming convention (ASSOCIATED PROCESS-TYPE-NAME) is mandated, with a "-a" suffix for released versions. Records must be retained for at least 10 years after the device's lifecycle ends.

A4: Corrective and Preventive Action (CAPA)

The CAPA process is triggered by non-conformities detected during bench, ux, clinical evaluation or complaints during post market surveillance. The QMO coordinates a "Five Whys" root cause analysis with technical leads. Before closing a CAPA, proof of implementation and an effectiveness review are required and should be documented..

A5: Equipment Disposal and Decommissioning

Obtaining user department assent and documenting the rationale for disposal is mandatory. High-risk devices (e.g., lasers, defibrillators) must be permanently disabled before disposal to prevent injury.

The Product Manager must update the Medical Device List with the final decommissioning date.

A6 & A6.1: Regulatory Monitoring & Consumer Compliance

The QMO maintains a List of Regulatory Requirements that is reviewed at least annually. A "Shift Right" policy integrates EMC (NTC), Cybersecurity (DICT), and Data Privacy (NPC) benchmarks directly into the design phase. Projects involving IoT or wireless transmission must pass Mandatory Type Approval.

A7: Personnel and Contractor Auditing

Core processes are audited annually to evaluate compliance and performance. Findings are categorized as Major Nonconformities (MNC+), Minor Nonconformities (MNC-), or Recommendations (REC). Auditors must remain objective and impartial, with findings reported during Management Review.

A8: Management Review

Top management conducts a formal review of KPIs, audit results, and CAPA status at least once per year. The primary output is a Management Review Report that assesses QMS efficacy and defines resource allocation for the next period.

SECTION B: HUMAN TESTING (CLINICAL EVALUATION)

B0: Preparation for Deployment

Requirements include the Pre-Clinical Evaluation Report to prove a balanced risk-benefit ratio. A Usability Evaluation Plan must be established to outline both formative and summative testing. Labels must include a Unique Device Identifier (UDI) for traceability.

B1: Contingency Planning during Deployment

Failure protocols require immediate removal of faulty devices from service and generation of a fault report. For participant safety, devices must feature physical and software-based emergency stop mechanisms. Hard-coded software limits prevent pain or over exertion. Supply and equipment contingencies must also be planned for the purpose of deployment for evaluation purposes such as n+1 equipment requirement or localized repair and support systems.

B2: Usability Testing Process and Procedure

Summative Usability Evaluation requires at least five participants per user group. Tests cover identified Hazard-Related Use Scenarios such as microshock or joint pain. Moderators record metrics like "Time to Complete" and "Give-up Rate" to validate safety.

B3: Process for Entering Human Trials

Human trials require Institutional Ethics Committee (IEC) approval and notification to regulatory bodies like the Philippine FDA . A Clinical Evaluation Plan (CEP) is developed, often involving external partners to reduce bias. Successful trials result in a Clinical Evaluation Report (CER) necessary for final market release.

B4: Clinical Readiness Level (CRL)

The CRL scale uses nine stages to communicate clinical progress to stakeholders. It moves from Phase 1: Conceptualization (CRL 1-2) to Phase 4: Market Adoption (CRL 8-9). This assessment tool is used alongside Technology Readiness Levels (TRL).

SECTION C: MARKET ENTRY AND POST-MARKET SURVEILLANCE

C0: Financial Valuation

Valuation integrates Venture Capital (VC) and Discounted Cash Flow (DCF) methods. These calculations are adjusted by risk-adjusted beta coefficients covering technology maturity and implementation feasibility. Decisions also incorporate the Total Cost of Ownership (TCO).

C1: Hazards Alerts and Safety Recalls

An inventory audit is triggered immediately upon receipt of a recall notification. Affected devices are withdrawn from service and labeled to warn against use. Following modifications, devices must undergo verification testing before returning to clinical use.

C2: Feedback Management and Customer Complaints

Incoming feedback is classified as Support Inquiry, Change Request, Complaint, or Health Risk. Health-related issues must be reported to the PRRC on the same day. Automated tracking is handled via the OS Ticket system.

C3: Procurement and Tendering

A committee review evaluates major acquisitions for suitability, compatibility, and Total Cost of Ownership. Suppliers must provide comprehensive technical documentation, including circuit diagrams and maintenance instructions. Standardization is encouraged to reduce training and support costs.

C4: Technical Infrastructure and Ticketing

The framework recommends a Linux-based (CentOS/LEMP) infrastructure for stability and granular security. Key components include Kanboard for inventory, OS Ticket for problem resolution, and Samba for secure record storage. All access should be secured using SSL encryption.

Work Instruction: Integrated Medical Device Lifecycle Management (ISOXIAN Framework)

1. Document Control Metadata (ISO 7200 Format) Field Details

Title: Work Instruction: ISOXIAN Framework (A0-C4)
Document Owner: Voltaire B. Dupo
Legal/Regulatory Basis: ISO 13485, PH FDA AO 2020-0010, RA 10173 (DPA), IEC 62366, ISO 14971
Status: For Review and Evaluation Released (Suffix: -a)
Initial Compilation Date: May 22, 2024 Language:English (US)

2. Project Initiation & Administrative Setup (A0 - A1)

2.1 Leadership & Roles

Core project roles must be appointed upon the management decision to initiate a device lifecycle. Roles are defined by independence and regulatory accountability:

- Product Manager: Coordinates the technical documentation compilation, maintains the Medical Devices List, manages Unique Device Identifier (UDI) registration, and initiates change requests. Reports to Management.
- Quality Management Officer (QMO): Documents QMS processes, maintains the perennial audit program, tracks CAPA effectiveness, and coordinates annual management reviews. Reports directly to Top Management.
- Person Responsible for Regulatory Compliance (PRRC): Reviews and releases technical documentation, ensures the EU Declaration of Conformity remains current, and oversees vigilance reporting.
- MANDATORY INDEPENDENCE: The PRRC must not be subjected to management instructions while carrying out regulatory duties to ensure objective safety oversight.

2.2 Hiring Workflow

- Staffing for research projects at DLSU follows the External Grants Management Office (EGMO) 6-step workflow:
- Contract Assignment: Assign a unique Control Number (e.g., E-2425T1001-01) to the research contract.
- Personnel Classification: Distinguish between Contract of Service (COS) for research staff and Consultancy Agreements for faculty.
- Mandatory Documentation: Collect the Personal Data Sheet (PDS), Terms of Reference (TOR), and mandatory tax forms (BIR Form 1901/1903).
- Justification: Submit formal justification letters for any late submissions to the workflow.
- Review: Internal administrative review of contract terms and scope of work.
- Approval: Formal sign-off and onboarding.

2.3 Kanban System Configuration

Configure the project board (e.g., Kanboard)

- For Research and Development there should be
    - Columns:
        - Management
            - Project Objectives
            - Target Market
            - Pain Point Statements
            - Estimation of Target Market Size
            - Filing and Approval of R&D Cycle and Funding Grant
        - Research and Development Activities to resolve Project Objectives
        - Documentation Development
            - User Manual
            - Design Manufacturing Record
            - Software/Firmware Audit
            - Investigators Brochure
        - Test Evaluation
            - Internal Laboratory Test Bench Protocol - Pre-compliance tests
            - PSL Compliance
            - CISPR Compliance
            - MISRA C++ Compliance
            - User Experience (UX) Testing
            - Clinical Protocol Development based on Identified Target Market
        - Change Management
            - Corrective Action Preventive Action
                - Issued with file reference naming
                - CAPA -<Project>-<Year> - <Number>

        - Closed Issued CAPA Items with Validation Documentation of resolution.
    - New Features and New Purpose of Product
        - Change Management Input Document that details the rationale for an alternative use for the device.
            - Setting of New Objectives for Clinical Validation of the new use case
            - Target Market
            - Pain Point Statements
            - Estimation of Target Market Size
            - Funding Grant Proposal Filling
            - Upon Approval of R&D Cycle and Funding Grant
                - Prompts a new Investigators Brochure
                - If new features and revised prototype is needed then
                - CISPR & PSL Compliance Redone
                - Software/Firmware Evaluation using MISRA C++ to check the new purpose
                - New User Manual that adds and discusses new features and use cases.
                - UX Testing of Interface
                - Clinical Protocol Development
- For the deployment stage this is done with the following seven standard columns and departmental swimlanes:
    - Columns:
        - 1. Pre-Purchase,
        - 2. Purchase,
        - 3. Commissioning,
        - 4. Equipment In Use,
        - 5. Preventive Maintenance,
        - 6. Corrective Maintenance,
        - 7. Decommissioning.
    - Swimlanes: The board must be organized into Back Office, Production, and Test & Evaluation swimlanes to ensure departmental traceability.

3. Engineering, Design & Document Control (A2 - A3)

3.1 Software and Hardware Realization

Engineering follows an evolutionary strategy integrating software and hardware development:
Software Lifecycle: Development must follow the sequence:
Architecture Design -> Coding -> Unit Testing -> Integration -> Verification -> Validation.

- All software products must follow a MAJOR.MINOR.PATCH versioning strategy.
- Hardware Tasks: Parallel to software, the engineering phase must document: Power Supply Evaluation, Processor Assembly, and Mechanical/Electronics Section Assembly.

3.2 Document Naming, Storage & Periodic Review

All documentation must adhere to the ASSOCIATED PROCESS-TYPE-NAME schema.

- Folder Structure: Documents move through Drafts (creation), Under Review (initialing/suggesting), and Released (finalized).
- IMPORTANT: All documents in the "Released" folder must be appended with a "-a" suffix (e.g., QMS-SOP-Audit-a). This suffix is mandatory for identifying the active, approved version of any document.

Periodic Review Cycles: Released QMS documents must be checked on a risk-based cycle:

- Category A (New/Updated): Review every 6 months.
- Category B (Stable): Review every 1 year.
- Category C (Long-term): Review every 3 years.

3.3 Retention Policy

- QMS Records: Retained for 10 years after the archival date.
- Technical Documentation: Retain for 10 years after the device's functional end-of-life.

4. Consumer Compliance & Design Direction (A6.1)

4.1 "Shift Right" Policy & EPDC Partnership

Compliance must be integrated during the design phase to prevent post-development market rejection which is more costly to correct.

EPDC Mandate:

To ensure circuit and board design compliance, the projects must enter into a partnership agreement with the Electronics Product Development Center (EPDC) for board layout and case translation. This allows already present considerations for compliance testing to be part of the generated circuit boards and cases.

Engaging with EPDC for final pre-commercialization tests for CISPR 11, CISPR 14 and PSL Compliance is also mandated by this standard. This allows the project outputs to pass the requirements set by the DTI thru the BPS, NTC and FDA Guidelines.

4.2 Technical Benchmarks

Agency	Standard/Benchmark	Key Requirements
NTC (RF/EMC)	CISPR 11/14/32/35	Mandatory Type Approval; RF/EM verification; Class A or B designation.
DICT (Cybersecurity)	NCSP 2023-2028; RA 1225	Vulnerability management and data encryption for all IoT/wired devices.
NPC (Data Privacy)	RA 10173 (DPA)	Privacy by design; unauthorized processing carries penalties of P4M fine and 6 years prison.
USA FDA	CFR 820	Medical Software

5. Risk Management & CAPA (A2, A4)

5.1 Risk Hierarchy (ISO 14971)

Total Number of Working Hours for the device Life Span must be estimated.

Mitigation must address hazards such as environmental hazards caused by the machine (H1), electrical hazard (H2), Physiological Hazards (H3), Energy Hazards (H4), Ergonomic (H5), and Human Behaviour Related Hazards (H6).

All hazards listed out are estimated in terms of effect on human that use the machine this is classified as Negligible (Hazard becoming an active threat to create an accident means no injury), Minor (effects are treated with first aid and settled), Severe (effects require hospitalization but not life threatening), Critical (effects are deadly).

An estimate of the duration of time where this hazard becomes present for the whole life is estimated. This builds a probability of occurrence value.

Probability is plotted versus severity of hazards occurrence. Hazards in the higher probability and severity areas are marked for addressing as first on the list. Addressing them means going through three possible processes..

- Inherently safe design: Eliminate the hazard (e.g., electrical isolation to prevent microshock).
- Protective measures: Automated alarms or physical guards, use of PPE, Utilization of software guards
- Information for safety: Warnings in manuals and user training.

5.2. 5-Step Corrective Action Protective Action (CAPA) Process

- Trigger/Input: Identify non-conformities from audits, complaints, or surveillance.
- Decision & Immediate Action: QMO and PRRC determine if field safety corrective actions are required.
- Root Cause Analysis (RCA): Must use the "Five Whys" method to determine the underlying failure.

- Implementation: Execute actions while verifying they do not negatively impact safety or regulatory compliance.
- Verification of Effectiveness: Document proof of implementation and confirm the issue is resolved.

6. Usability & Clinical Evaluation (B0 - B4)

6.1 Usability Testing (IEC 62366)

- Summative Evaluation requires at least 3-5 participants per user group:
- User Groups Must include
    - Engineers/Technicians: Maintenance and setup.
    - Allied Health Professionals and Physicians
    - Target demographic (e.g., Stroke patients aged 54+).

6.2 Human Clinical Trials Pathway

Transition from Pre-Clinical Evaluation to the Clinical Evaluation Plan (CEP) must comply with PH FDA AO 2020-0010. External clinical partners should be engaged to minimize bias in the Clinical Evaluation Report (CER).

6.3 Clinical Readiness Level (CRL)

Assess progress using the 9-stage CRL scale:

- Phase 1: Conceptualization (CRL 1-2): Clinical competence and need identification.
- Phase 2: Concept Validation (CRL 3-4): Prototype testing and clinical environment user studies.
- Phase 3: Product Development (CRL 5-7): Animal studies (IRB approved) and randomized controlled trials. Some countries have stopped doing animal testing since human tests usually show differences for the data gathered.
- Phase 4: Market Adoption (CRL 8-9): Usability validation and post-market clinical follow-up.

7. Deployment, Feedback & Surveillance (B1, C1, C2)

7.1 Contingency Planning

- Device Replacement and Servicing when deployed for Clinical Studies.
- Process and Reporting Systems for treatment of injuries related to failure during device use.
- User Manual Recorded items related to:
    - Safety Mechanisms: Devices must feature physical and software-based Emergency Stop mechanisms.
    - Hard-coded Limits: To prevent over-exertion, software must include hard-coded limits (e.g., max 15 repetitions or maximum exposure values).

7.2 Feedback & Severity Management

Incoming feedback via OS Ticket must be synthesized into Severity Classifications:

- Support Inquiry (Low Severity): General help and training.
- Change Request (Low/Medium Severity): Functionality modifications.
- Customer Complaint (Medium Severity): Technical bugs or non-safety non-conformities.
- Health Impact Issue (High Severity): Incidents causing death or serious deterioration of health.

7.3 Vigilance Reporting

- Internal: "Same Day" escalation to the PRRC for all High Severity (Health Impact) issues.
- External: Serious incidents involving death or serious health deterioration must be reported to the authorities within a 2 to 15-day timeline, depending on the severity and manufacturer advice.

8. Market, Procurement & Infrastructure (C0, C3, C4)

8.1 Financial Valuation & TCO

- Valuation: Use the Venture Capital (VC) or Discounted Cash Flow (DCF) methods with Risk-Adjusted Beta Coefficients (Technology Maturity, Feasibility, Competence).
- Total Cost of Ownership (TCO): Evaluated by committee.

Formula: TCO = Purchase Price + Sum of Repair Costs + Decommissioning Cost.

8.2 Technical Stack

- Infrastructure must be LEMP/Linux-based (Linux, Nginx, MariaDB, PHP) for stability and security.
- Samba: Central repository for sharing with permission-restricted access.
- Webmin: Remote server administration.
- Encryption: Encrypted SSL is mandatory for any network connected processes.

9. Internal Audit & Management Review (A7 - A8)

9.1 Audit Categorization

Findings are classified as:

- Major Nonconformities (MNC+): Systematic failures indicating the QMS cannot deliver safety.
- Minor Nonconformities (MNC-): Isolated events (e.g., single mislabeled record).
- Recommendations (REC): Suggestions for process optimization.

9.2 Management Review Inputs

Mandatory annual assessment of: KPI trends, audit results, CAPA status, feedback/complaints, Risk Management/PMS updates, and regulatory status.

10. Equipment Decommissioning & Disposal (A5)

10.1 Protocol for Disposal

- User Assent: Obtain formal agreement from the user department.
- Rationale: Document the reason (obsolescence, unrepairable fault, or safety).
- Notification: Formally notify asset management for removal from financial records.
- Traceability: Move the Kanban card to "Decommissioning" and update the DHR.

10.2 High-Risk Safety

- Permanent Disabling: High-risk units (lasers, defibrillators, diathermy generators) must be permanently disabled prior to disposal.
- Hazardous Materials: Ensure regulated disposal of batteries, heavy metals (lead, mercury, beryllium), and pressure vessels.

Internal Audit Checklist: ISOXIAN Compliance

Field	Information
Document Title	Internal Audit Checklist: ISOXIAN Compliance
Document Number	ISOX-AUD-2026-002
Legal Owner	De La Salle University (DLSU)
Created By	Quality Management Officer (QMO)
Approved By	Management (CEO/Director)
Audit Date	Insert Date
Auditee(s)	Insert Personnel Names
Auditor(s)	[Insert Auditor Name - Must be objective/impartial 1, 3]

Section	Clause / Requirement	Yes	No	Comments
A0	Is there a designated PRRC who operates independently of management instructions? 4			
A1	Is the project tracked via a 7-column Kanban (Pre-Purchase to Decommissioning)? 5, 6			
A1	Has the EGMO 6-step workflow and unique Control Number been used for all personnel? 7, 8			
A2	Are the Software Development Plan (SDMP) and SOUP List documented and current? 9, 10			
A3	Do all released documents use the ASSOCIATED PROCESS-TYPE-NA			

	ME naming schema with a "-a" suffix? 11, 12			
A3	Is there evidence of a 10-year retention policy for technical documentation? 13, 14			
A4	Does the CAPA process utilize the "Five Whys" method for root cause analysis? 14, 15			
A5	Is user department assent documented prior to equipment disposal? 14, 16			
A5	Were high-risk devices (e.g., lasers) permanently disabled before decommissioning? 14, 17			
A6	Has the List of Regulatory Requirements been reviewed within the last 12 months? 18, 19			
A6.1	Is there evidence of "Shift Right" policy integration (NTC/DICT/NPC benchmarks)? 20, 21			
A8	Was a Management Review conducted annually covering KPIs, Audits, and CAPAs? 22, 23			
B0	Does the Clinical Evaluation Plan (CEP) include a balanced risk-benefit ratio? 24, 25			

B1	Are emergency stop mechanisms (physical and software) verified for deployment? 26			
B2	Did the Summative Usability Evaluation include at least 5 participants per user group? 14, 27			
B3	Is there evidence of Institutional Ethics Committee (IEC) or PH FDA AO 2020-0010 compliance? 14, 28			
B4	Is the Clinical Readiness Level (CRL) assessed and reported alongside TRL? 29, 30			
C1	In the event of a recall, was an inventory audit and immediate withdrawal performed? 31, 32			
C2	Are health-impacting feedback issues escalated to the PRRC on the same day? 14, 33			
C3	Does the procurement process evaluate Total Cost of Ownership (TCO)? 14, 34			
C4	Is the infrastructure secured via Linux (LEMP) with Samba granular permissions and SSL? 35-37			

The process for generating technical, research, and regulatory documentation under the ISOXIAN Framework is a multidisciplinary workflow that integrates engineering realization (A2), quality management (A3-A4), and usability validation (B0-B2) to maintain a "Golden Thread" of traceability.

1. Generating the Investigator's Brochure (IB)

The IB serves as a comprehensive summary of the non-clinical and clinical data on the investigational product. Its generation relies on inputs from the following ISOXIAN sections:

- Roles and Affiliations (A0): Populates the list of authors and their institutional affiliations.
- Regulatory Classification (A1): Provides the rationale for classifying the device (e.g., Class B) and its intended use.
- Non-Clinical Study Data (A2, A7): Incorporates movement rate data for digits and the wrist, along with the Medical Robot Integrated Compliance Checklist.
- Document Control (A3): Maintains the IB's revision history and effective dates.
- Risk Analysis (A4, B1): Details the technical risk analysis, residual risks, and the Hazard Occurrence Probability Matrix.
- Clinical Readiness (B0, B2): Includes the rationale for the robotic orthosis and medical spasticity scale data (MAS).

2. Generating the User Manual

The User Manual provides the operational instructions for clinicians and patients and serves as a vital input for the subsequent UX Test Protocol.

- Personnel Contacts (A0, A8): Identifies the clinical points of contact and technical support information.
- Device Description (A1): Establishes the general purpose and description of the device.
- Logistics and Maintenance (A2, A6): Provides instructions for installation, handling, storage, transport, and sterilization.
- Operational States (B0, B1): Defines the specific exercise modes (0–4) and the emergency stop procedures for both knob and button interfaces.
- Pre-use Verification (A7): Outlines the Self-Test procedure that occurs upon power-on to ensure the system is safe for use.
- Troubleshooting (A4): Lists common concerns and their solutions based on previous design feedback.

3. Software Audit Process

The software audit evaluates the APRO firmware against US FDA 21 CFR Part 820 and IEC 62304 to ensure medical-grade compliance.

- Requirements Mapping (A1): Auditors map functional requirements (FR) and safety requirements (SR) to the Software Requirements Specification (SRS).
- Architectural Review (A2): Verifies the use of non-blocking timing logic (millis) and deterministic state machines to prevent "blind" durations during operation.
- Risk Mitigation Audit (A4, B1): Confirms the implementation of the Hardware Watchdog Timer (WDT) and evaluates E-Stop Interrupt (ISR) latency to ensure immediate execution regardless of software state.
- Traceability and Verification (A3, A7): Auditors utilize a Traceability Matrix to link every safety requirement to specific code implementation and verify results through the Verification Test Protocol (VTP).
- Boundary Value Analysis (B2): Tests the software's hardcoded limits, such as ensuring the system transitions to a "Limit Reached" state exactly at 15 repetitions.
- Audit Conclusion (A8): The process culminates in a Final Verification Report (FVR) and a Statement of Conformity signed by the Lead Engineer and QA Lead.

To generate the technical and regulatory documents for the APRO project, the team utilizes a structured workflow that integrates management (A0-A1), engineering (A2-A3), and clinical/usability (B0-B3) processes.

4. Design Master Record (DMR)

The DMR serves as the definitive manual for manufacturing the device, ensuring reproducibility under ISO 13485 standards.

Ownership and Compilation (A0): The Product Manager is responsible for coordinating the compilation of technical documentation, while specialized engineering leads provide the technical inputs.

- Technical Specifications (A1, A2): The process begins with determining the Device Classification (e.g., Class IIa) and intended use, which dictates the rigor of the design.

- Bill of Materials (A2, C3): The Software and Hardware Engineers validate components for the 5-motor and 7-motor variants, sourcing from approved vendors like Jassen Industries.
- Manufacturing Instructions (A2, A6.1): Detailed electronic and mechanical assembly steps are documented, including Processor Timing requirements and Cybersecurity specifications such as encryption protocols.
- Quality Assurance Checklists (A7): The DMR integrates a Machine Inspection Checklist based on the AS/NZS 3551 standard, covering electrical safety and mechanical verification, such as the hard-coded 15-repetition software limit to prevent joint pain.

5. Generating the Design Transfer Checklist

This document facilitates the formal transition from Research and Development (R&D) to manufacturing.

- Specification Transfer (A2, A3): The Quality Management Officer (QMO) ensures all R&D "ideas" are now "frozen" engineering documents, including validated BOMs, finalized 3D CAD files, and version-controlled software code.
- Process Documentation (A2): It verifies that the Production Flow Chart and specific Work Instructions (WI) for tasks like servo motor testing and glove assembly are documented in the QMS Manufacturing Guide.
- Verification and Validation (V&V) (B2): The checklist confirms that the device meets original design intent by verifying results from the Product Test Protocol, including spastic condition simulations and calibration status.
- Approval Sign-Off (A0, A8): Final transfer requires sign-off from the R&D Lead (DHF Owner), the Manufacturing Lead (DMR Owner), and the QMO after a formal Design Transfer Review meeting.

6. Application for Ethics Review

This application (Project Code: 2025-090) is a prerequisite for human testing and is governed by institutional ethics policies.

- Study Rationale and Objectives (A1): The Project Leader defines the rationale based on clinical gaps—such as the need for early finger and wrist movement post-surgery—and sets objectives focused on User Experience (UX) and Value Engineering (VE).

- Methodology and Design (B0, B2): The application adopts the 5 E's approach (Effective, Efficient, Engaging, Error Tolerant, Easy to Learn) and defines three participant groups: Group A (Allied Health Professionals), Group B (Seniors 54+), and Group C (Engineers).
- Risk and Safety Management (B1): It includes a detailed Risk Management Plan identifying hazards like discomfort in the knuckles or exposure to non-ionizing radiation, and outlines mitigations such as limited exposure time (2 minutes) and mandatory handwashing hygiene.
- Data Handling and Privacy (A3): The application defines a Data Management Plan ensuring compliance with the Philippine Data Privacy Act (RA 10173), including quasi-anonymity, encryption, and a 1-year data retention/destruction policy.
- Submission Checklist (B0, B3): The final application package requires the Investigator's Brochure (IB), User Manual, and Informed Consent Forms as supplementary evidence.